

# TERRA TRACE

FORENSIC AUDIT ENCLAVE & DECISION-SUPPORT SYSTEM

## Team Presentation Guide, Real-World Case Studies & Simplified Math Playbook

**NON-PUNITIVE CORE RULE:** Terra Trace NEVER cancels an exam or disqualifies a candidate automatically. It acts strictly as an evidentiary decision-support enclave: every mathematical anomaly is flagged with clear physical evidence and handed over to human judicial and administrative officers for statutory adjudication.

### 1. The 5-Minute Pitch Script & Click Flow (For Non-Tech Presenters)

**YOUR 30-SECOND OPENING HOOK (Say this word for word):**

"Every major exam integrity scandal in India takes months to investigate because physical OMR paper sheets, computer server databases, and exam hall seating plans live in separate departmental silos. **Terra Trace unites all three** into an air-gapped forensic decision register that catches fraud in seconds with mathematical legal proof ready for a court of law."

#### The Exact 5-Click Presentation Routine:

**Click 1 (Tab 01 - Ingestion):** Click Load WBSSC Case (Real).

*What you say:* 'Notice that before any math runs, all three files are stamped with SHA-256 cryptographic fingerprints. This guarantees chain-of-custody under Section 65B of the Evidence Act so no one can accuse auditors of tampering with evidence.'

**Click 2 (Tab 01 - Raw Inspector):** Click [Watch Raw Datasets ▼].

*What you say:* 'Look at the crimson cells in server.csv. Candidate WBSSC\_SLST\_0001 got only 4 questions right on paper (raw score 4.0), but someone altered their database record to 53.0 marks—an unauthorized jump of +49 marks to push them over the qualifying cutoff.'

**Click 3 (Tab 02 - Analytical Engine):** Show the 3-layer formula (45% / 35% / 20%).

*What you say:* 'We do not use black-box AI or ChatGPT that hallucinates. We use three deterministic mathematical layers: reconciliation discrepancies, centre bell-curve divergences, and desk-neighbour collusion.'

**Click 4 (Tab 03 - Triage & Drilldown):** Click on candidate WBSSC\_SLST\_0001.

*What you say:* 'Here is the full forensic dossier: question-by-question physical bubble responses, the centre's distribution bell curve, and an interactive 5x5 seating grid showing their desk neighbours.'

**Click 5 (Tab 03 - PDF & Adjudication):** Click Export PDF Dossier, then click Adjudicate.

*What you say:* 'With one click, we generate a court-admissible Section 65B PDF evidence dossier. The officer marks the case as Confirmed with a mandatory written justification. Zero automated disqualifications.'

### 2. Real-World Case Studies Explained in Plain English

#### Case 1: The WBSSC 2016 Teacher Recruitment Scam (Calcutta High Court)

**What happened in real life:** Thousands of candidates handed in essentially blank OMR sheets, scoring single-digit marks (2 to 4 marks out of 60). Inside officials manually edited the MySQL database server records and published inflated scores of 51 to 54 marks so these candidates qualified for government school teaching jobs.

**How Terra Trace catches it: Layer 1 (The Receipt vs Bank Account Check).** Just like comparing your grocery receipt with your bank debit statement, Terra Trace compares the raw optical bubble scan with the server table. Any jump  $\Delta$ ge +10\$ marks triggers an immediate red flag objection.

## Case 2: The NEET-UG 2024 Controversy (Haryana & Rajkot Centres)

**What happened in real life:** In NEET 2024, an unprecedented 67 candidates secured All India Rank 1 with a perfect 720/720 score. At a single test centre in Jhajjar, Haryana (Hardayal Public School), 6 students scored 720, and others scored 718 and 719 (which is mathematically impossible on a +4/-1 marking scheme without arbitrary grace marks). In Rajkot, Gujarat, scores were abnormally clustered.

**How Terra Trace catches it: Layer 2 (The Too-Many-Geniuses Bell Curve Check).** In any normal exam, marks form a natural bell curve (a few low, most in the middle, very few top scores). When an exam paper leaks or answers are broadcast, the curve deforms into an unnatural 'shark fin' spike. Terra Trace's Kolmogorov-Smirnov test flags this centre-wide anomaly ( $\$D > 0.35$ ,  $p < 0.001\$$ ) automatically.

## 3. What is "Synthetic Data" and Why Do We Use It?

**In plain English:** Synthetic data is computer-generated test exam data that mimics real students, questions, and marks.

**Why is it crucial?**

- **Student Privacy (DPDP Act):** Real exam bodies cannot legally publish millions of students' personal Aadhaar numbers, phone numbers, and addresses. Synthetic data lets us demonstrate the software publicly without leaking real student records.
- **Edge-Case Testing:** We can simulate extreme scenarios—like two desk neighbours deliberately copying each other's wrong answers—to prove our math catches it.
- **Proving Zero False Positives:** When we run a 'Clean Control' synthetic cohort (honest students), Terra Trace flags **exactly zero candidates**, proving to judges that honest students are never falsely accused.

## 4. The Math Explained (Analogies for Non-Technical Teammates)

$$\text{CombinedRisk} = (0.45 \times \text{Reconciliation}) + (0.35 \times \text{Macro}) + (0.20 \times \text{Micro})$$

### • Layer 1: Score Reconciliation (45% Weight) — 'Did someone hack the database?'

*Analogy:* You have a physical paper receipt saying you bought Rs. 4 worth of items. But the shop owner's computer says you paid Rs. 53. The difference  $\Delta = +49\$$  is caught instantly. We assign it the highest weight (45%) because physical paper bubbles cannot lie.

### • Layer 2: Macro Centre Patterns (35% Weight) — 'Is the whole exam centre cheating?'

*Analogy:* If a random classroom in a small town suddenly has 15 national toppers while top schools in Delhi and Mumbai have none, something is fishy. The Kolmogorov-Smirnov (KS) test compares the centre's score curve against the state curve. If the curve is deformed, the entire centre is flagged.

### • Layer 3: Micro Seating Collusion (20% Weight) — 'Did desk neighbours copy each other?'

*Analogy:* If two students both get question #1 right, that is normal. But if two students sitting right next to each other both pick the exact same bizarre, incorrect answer (like option 'D' when the real answer is 'A') across 8 different questions, that is not a coincidence—it is collusion. The Wollack  $\omega$  index measures this probability.

## 5. Defense Q&A;: Handling Tough Evaluator Questions

### Q: Why not use AI / Machine Learning?

A: "In court, 'black box' machine learning weights and ChatGPT hallucinations are legally inadmissible. Terra Trace uses deterministic statistical math. Every score is explainable down to the exact formula."

### Q: Can an auditor tamper with evidence to frame someone?

A: "No. The second raw files are uploaded, their SHA-256 block hashes are stamped immutably. Any manual edit to a CSV alters the hash and instantly invalidates the Section 65B legal certificate."

**Q: What if a centre scored well because of great teachers?**

*A: "Terra Trace does not flag centres for high average marks. It tests for unnatural distribution shapes—like an artificial spike strictly at the qualifying cutoff with zero room variance. Normal high-performing schools exhibit natural bell curves."*

**Q: How is student privacy protected?**

*A: "Terra Trace runs air-gapped on on-premise government servers without internet. Student names, phone numbers, and Aadhaar numbers are never ingested—only anonymized Roll IDs and Seat Coordinates."*